When $n = 1,\ a_1^2 + 2a_1b_1 - b_1^2 = 1 + 4 - 4 = 1$
Assume $a_k^2 + 2a_kb_k - b_k^2 = 1$ is true

$$\begin{aligned} a_{k+1}^2 + 2a_{k+1}b_{k+1} - b_{k+1}^2 &= (a_k + 2b_k)^2 + 2(a_k + 2b_k)(2a_k + 5b_k) - (2a_k + 5b_k)^2 \\ &= a_k^2 + 2a_kb_k - b_k^2 \\ &= 1 \end{aligned}$$

By mathematical induction, $a_n^2 + 2a_nb_n - b_n^2 = 1$ is true for all $n \geq 1$.

(i)

$$b_{n+1} = 2a_n + 5b_n \geq 5b_n$$

$$b_1 = 2 \implies b_n \geq 2 \cdot 5^{n-1}$$

Dividing $(*)$ by $b_n^2 \implies c_n^2 + 2c_n - 1 = \dfrac{1}{b_n^2}$

$$b_n \geq 2 \cdot 5^{n-1} \implies \lim_{n\to\infty} \frac{1}{b_n^2} = 0$$

$$(c_n + 1)^2 = 2 + \frac{1}{b_n^2}$$

$$c_n + 1 > 0 \implies c_n + 1 = \sqrt{2 + \frac{1}{b_n^2}}$$

$$\lim_{n\to\infty} (c_n + 1) = \lim_{n\to\infty} \sqrt{2 + \frac{1}{b_n^2}} = \sqrt{2}$$

$$\lim_{n\to\infty} c_n = \sqrt{2} - 1$$

(ii)

$$c_n^2 + 2c_n - 1 = \frac{1}{b_n^2} > 0$$

$$(c_n - (\sqrt{2} - 1))(c_n + \sqrt{2} + 1) > 0$$

$$c_n + \sqrt{2} + 1 > 0 \implies c_n > \sqrt{2} - 1$$

$$c_n + 1 > \sqrt{2} \implies \frac{2}{c_n + 1} < \frac{2}{\sqrt{2}} = \sqrt{2}$$

$$\therefore \quad \frac{2}{c_n + 1} < \sqrt{2} < c_n + 1$$

$$a_2 = 1 + 2(2) = 5,\ b_2 = 2(1) + 5(2) = 12$$

$$a_3 = 5 + 2(12) = 29,\ b_3 = 2(5) + 5(12) = 70$$

$$\implies c_3 = \frac{29}{70}$$

$$\frac{2}{c_3 + 1} < \sqrt{2} < c_3 + 1$$

$$\frac{140}{99} < \sqrt{2} < \frac{99}{70}$$

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